

ARUN MADHUSUDHANAN

(857)-381-3654 | madhusudhanan.a@northeastern.edu | [Linkedin](#) | [Portfolio](#) | [Github](#) | Boston, MA

Availability: January 2025

EDUCATION

Master of Science in Robotics (Conc: Computer Science), Northeastern University, Boston, MA Dec 2024
Relevant Coursework: Computer Vision, Reinforcement Learning, AI, Robotics Sensing and Navigation
GPA: 4.0/4.0

Bachelor of Technology in Mechanical Engineering, NIT, Calicut, India. GPA: 8.89/10.0 2014 - 2018

SKILLS

Programming Skills	C++, Python
Libraries	PyTorch, Cuda, OpenCV, PCL, Open3D, TensorRT, ONNX, Scikit-learn, NumPy, Eigen
Software tools	MATLAB, ROS, Docker, Git, Ubuntu, AWS ECS, FastAPI, OriginPro, Solidworks, Abaqus (FEA)

EXPERIENCE

Machine Learning Research Intern

Festo, Boston, MA Jul 2023 - Dec 2023

- Developed a robust machine learning pipeline to predict the output parameters of a high-precision liquid dosing unit achieving an error rate of less than 2.5%; the device with incorporated ML model is currently under patent application.
- Managed all aspects of the ML pipeline, from dataset collection and feature extraction to data labeling, model development, and live inferencing.
- Evaluated various models (LSTMs, GRUs, Neural Networks and Decision Trees) to identify the most accurate model for predictions.
- Optimized software modules for sensor activation, hardware drivers, and data storage by refactoring code, fixing bugs, and implementing multi-threading, resulting in a significant reduction in system latency.

Graduate Teaching Assistant, Computer Vision

Northeastern University, Boston, MA Jan 2024 - April 2024

- Reviewed code, debugged issues, and graded projects in C++, Python, OpenCV, and PyTorch for a cohort of 120+ students.
- Mentored students on topics such as image processing, object recognition, camera calibration, keypoints & stereo vision.

Wells Engineer

ExxonMobil, India July 2018 – July 2022

- Led beta testing of company-wide tubular design software, ensuring robustness and reliability before release to users.
- Stewarded and improved the tubular design workflow for business divisions across the world in accordance with industry standard API 5C5, resulting in \$230k immediate savings and long-term synergistic benefits.

PROJECTS

Multi-Modal Open-Vocabulary 3D Indoor Scene Understanding Nov 2024 – Dec 2024

- Designed a multi-stage pipeline for open-vocabulary 3D scene graph construction, supporting audio, text, and image queries using Meta's ImageBind, for downstream planning in mobile robots.
- Leveraged foundation models (RAM, GroundingDINO, SAM, CLIP) to extract and encode semantic object features onto PointCloud data in the CLIP feature space from 2D images.
- Unified 3D object representations by merging object masks based on spatial-visual similarity and constructed hierarchical scene graphs to capture object-to-room relationships, enabling complex spatial queries.

Other Academic Projects

Dec 2022 – Oct 2024

- Developed a **3D Object Classification** pipeline using GRNet for point cloud completion and custom PointNet implementation, achieving 93.8% accuracy vs. PointNet++'s 70% on incomplete point clouds. [\[Code\]](#) [\[Report\]](#)
- Built a robust **Vehicle Tracking** system by fusing LiDAR and Radar data with Unscented Kalman Filter. [\[Code\]](#)
- Created **multi-sensor object tracking framework** using Camera and LiDAR to assess collision time in ego lane using independent TTC calculations. [\[Code\]](#)
- Built a **Structure from Motion** pipeline in C++ with Bundle Adjustment for 3D reconstruction from images. [\[Code\]](#)
- Developed monocular **Visual Odometry** system in C++ featuring pose graph optimization using GTSAM and GPS integration for scale estimation. [\[Code\]](#)
- Optimized **YOLOv8** with Nvidia TensorRT, achieving real-time inference speeds of 60 FPS. [\[Code\]](#) [\[Report\]](#)
- Implemented **Deep Learning Models from Scratch**: Tiny-NeRF for novel view synthesis, PointNet for point cloud processing, Visual Transformer, GAN & VAE [\[Code\]](#)
- Developed a scalable **FastAPI Image Classification Application** using AWS ECS and Docker containerization [\[Code\]](#)